Media Mix Modelling

Measuring marketing effectiveness for a fashion retail client

DATA

117 weeks of sales, marketing & macro data

METHOD

OLS regression with adstock & log transforms

OBJECTIVE

Identify channel ROI & optimise budget allocation

Model Structure & Coefficients

VARIABLES INCLUDED

Variable	Group	Coeff
log_economy	Macro	+0.605 ***
log_confidence	Macro	+0.598 ***
log_unemployment	Macro	−0.437 ***
log_tourists	Macro	−0.076 **
log_online	Marketing	+0.015 **
log_adstock_press	Marketing	−0.003 *
christmas_dummy	Control	−0.117 ***
public_holidays	Control	−0.020 **
weather_index	Control	−0.001 *
log_competition	Control	+0.002 *
brand_knowledge	Control	+0.129 .
log_sales_lag1	Dynamic	+0.168 **

WHY THESE VARIABLES?

- **Macro variables** — economy, confidence & unemployment explain the largest share of sales variation.
- **Online** — TV, banners & radio were tested but insignificant. Online is the sole measurable marketing channel.
- **Adstock on press** — captures carry-over effect of advertising beyond the week of spend.
- **Sales lag** — absorbs week-to-week momentum, resolving autocorrelation

Model Validation

0.896

Adjusted R^2

89.6% of sales variation explained

1.79

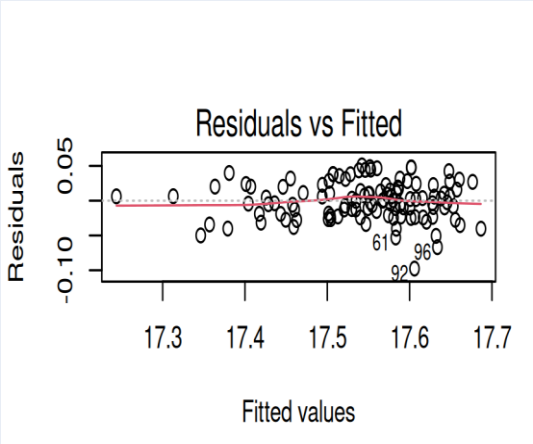
Durbin-Watson

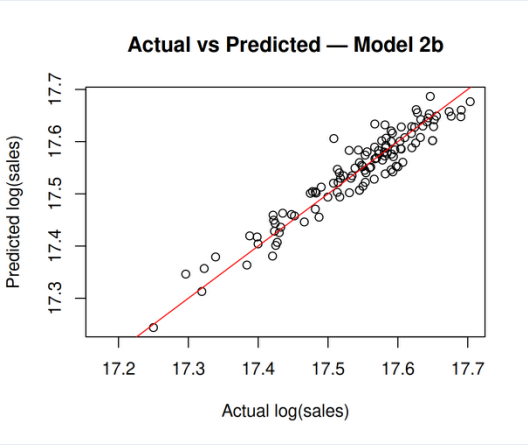
No meaningful autocorrelation

8.23

Max VIF

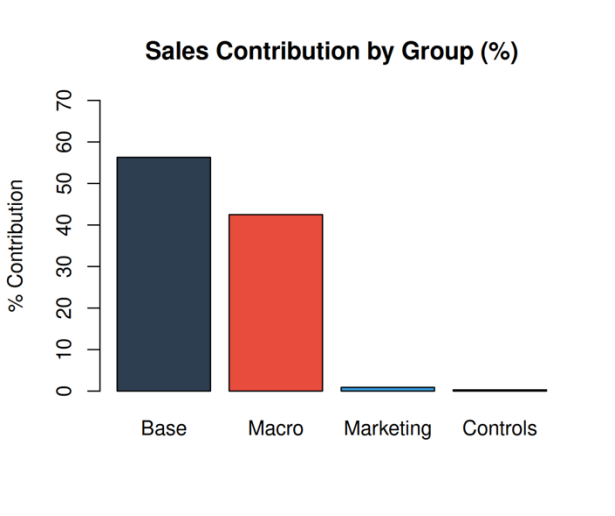
Below critical threshold of 10





What Drives Sales?

Macroeconomic conditions and structural brand strength explain 98.8% of sales — marketing contributes 0.9%



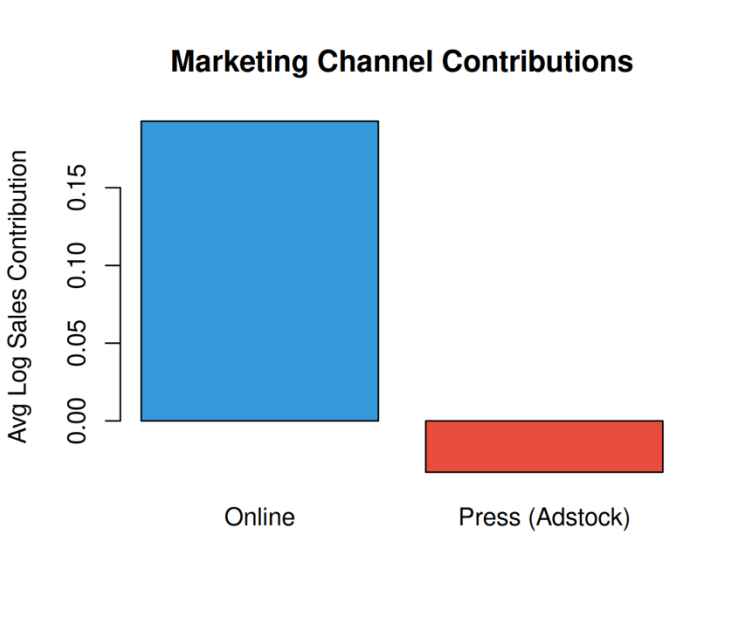
56.3% **Base**
Structural brand sales & weekly momentum

42.5% **Macro**
Economy, confidence, unemployment, tourists

0.9% **Marketing**
Online (positive)
Press (caveated)

0.3% **Controls**
Weather, holidays, brand, competition

Marketing Channel Performance & ROI



CHANNEL ROI SUMMARY

Channel	Annual Spend	Attr. Sales	ROI
Online	£29.9m	£11.9m	**0.40**
Press	£5.7m	−£2.0m	**−0.36**

KEY NOTES

- **Online ROI = 0.40** — Every £1 in online spend returns £0.40 in attributed sales. The only channel with a measurable positive return.
- **Press ROI caveated** — Negative coefficient reflects reverse causality (reactive spend during low-sales periods).
- **TV, banners & radio** — Could not be measured. Coefficients insignificant in all model iterations.

Budget Recommendations & Next Steps

↑ Increase Online

- Only channel with a measurable, positive ROI (0.40)
- Sole significant marketing variable in the final model
- Recommend increasing allocation from current £29.9m

? Investigate Press

- Negative coefficient is likely reverse causality, not a true effect
- Cannot confidently recommend cutting spend on this data alone
- Run a press blackout test to isolate the true effect

✦ Measure TV & Banners

- Both channels insignificant — could not be attributed
- Longer time series needed to isolate effects
- Do not reallocate budget away without further evidence